THE LOCUS WHERE A FINITE PRESENTATION MODULE IS FLAT IN DIMENSIONS AT LEAST N IS REPRESENTABLE BY A MONOMORPHISM OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a reduction of representability to a single etale cover, a descending induction on n through quasi-finite base change, and a complete devissage plus constructibility argument reducing to the flat case

Editorial note. The label theorem-flat-dimension-n-representable names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own theorem environment in the chapter source.

Editorial note. The counted statement is the author's theorem environment labelled theorem-flat-dimension-n-representable (2195-2203) and the counted proof is the proof environment beginning at 2205, so the pairing is by adjacency.

Editorial note. The proof is complete as the author wrote it: every reduction is justified where it is made and the only external input is the flat case quoted at the end, which is recorded among the allowed prerequisites.

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Allowed standard prerequisites (editorial).

- The functor classifying flatness in dimensions at least n, its sheaf property for the fppf topology, and the fact that it is a subfunctor of the corresponding functor with a larger bound
- Purity of a module relative to the base, its preservation under quasi-finite base change and under etale localization, and the constructibility of the relative assassin
- Etale descent for monomorphisms and for morphisms of finite presentation, the existence of a complete devissage in an etale neighbourhood, and the flat case of the statement

Source reference. Original lemma 02L0: sources/stacks-chow/tags.txt, source lines 3391–3391. Prerequisite pointer invoked at proof line(s) 2240; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02YK: sources/stacks-chow/tags.txt, source lines 3879–3879. Prerequisite pointer invoked at proof line(s) 2239; it names a standard

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fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02W5: sources/stacks-chow/tags.txt, source lines 3792–3792. Prerequisite pointer invoked at proof line(s) 2235; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05UF: sources/stacks-chow/tags.txt, source lines 7514–7514. Prerequisite pointer invoked at proof line(s) 2307; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05HU: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 640–640. Prerequisite pointer invoked at proof line(s) 2272; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05MV: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2122–2122. Prerequisite pointer invoked at proof line(s) 2207; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05UD: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2169–2169. Prerequisite pointer invoked at proof line(s) 2296; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0CWF: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2030–2030. Prerequisite pointer invoked at proof line(s) 2279; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00PP: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 525–525. Prerequisite pointer invoked at proof line(s) 2221; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 02W8: sources/stacks-chow/tags.txt, source lines 3795–3795. Prerequisite pointer invoked at proof line(s) 2237; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05LW: sources/stacks-chow/tags.txt, source lines 7250–7250. Prerequisite pointer invoked at proof line(s) 2292; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CT8: sources/stacks-chow/tags.txt, source lines 16058–16058. Prerequisite pointer invoked at proof line(s) 2233; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 05MT: sources/stacks-spaces-flat/pinned-original-source.tex, source lines 2085–2085. Prerequisite pointer invoked at proof line(s) 2197; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0CVV: sources/stacks-chow/tags.txt, source lines 16150–16150. Prerequisite pointer invoked at proof line(s) 2289; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Theorem 0.1. *In Situation 05MT. Assume moreover that f is of finite presentation, that $\mathcal{F}$ is an $\mathcal{O}_X$-module of finite presentation, and that $\mathcal{F}$ is pure relative to Y. Then F_n is an algebraic space and $F_n \to Y$ is a monomorphism of finite presentation.*

Proof. The functor F_n is a sheaf for the fppf topology by Lemma 05MV. Since $F_n \to Y$ is a monomorphism of sheaves on $(Sch/S)_{fppf}$ we see that $\Delta : F_n \to F_n \times F_n$ is the pullback of the diagonal $\Delta_Y : Y \to Y \times_S Y$. Hence the representability of Δ_Y implies the same thing for F_n. Therefore it suffices to prove that there exists a scheme W over S and a surjective étale morphism $W \to F_n$.

To construct $W \to F_n$ choose an étale covering $\{Y_i \to Y\}$ with Y_i a scheme. Let $X_i = X \times_Y Y_i$ and let $\mathcal{F}_i$ be the pullback of $\mathcal{F}$ to X_i. Then $\mathcal{F}_i$ is pure relative to Y_i either by definition or by Lemma 00PP. The other assumptions of the theorem are preserved as well. Finally, the restriction of F_n to Y_i is the functor F_n corresponding to $X_i \to Y_i$ and $\mathcal{F}_i$. Hence it suffices to show: Given $\mathcal{F}$ and $f : X \to Y$ as in the statement of the theorem where Y is a scheme, the functor F_n is representable by a scheme Z_n and $Z_n \to Y$ is a monomorphism of finite presentation.

Observe that a monomorphism of finite presentation is separated and quasi-finite (Morphisms, Lemma 0CT8). Hence combining Descent, Lemma 02W5, More on Morphisms, Lemma 02W8 , and Descent, Lemmas 02YK and 02L0 we see that the question is local for the étale topology on Y.

In particular the situation is local for the Zariski topology on Y and we may assume that Y is affine. In this case the dimension of the fibres of f is bounded above, hence we see that F_n is representable for n large enough. Thus we may use descending induction on n. Suppose that we know F_{n+1} is representable by a monomorphism $Z_{n+1} \to Y$ of finite presentation. Consider the base change $X_{n+1} = Z_{n+1} \times_Y X$ and the pullback $\mathcal{F}_{n+1}$ of $\mathcal{F}$ to X_{n+1}. The morphism $Z_{n+1} \to Y$ is quasi-finite as it is a monomorphism of finite presentation, hence Lemma 00PP implies that $\mathcal{F}_{n+1}$ is pure relative to Z_{n+1}. Since F_n is a subfunctor of F_{n+1} we conclude that in order to prove the result for F_n it suffices to prove the result for the corresponding functor for the situation $\mathcal{F}_{n+1}/X_{n+1}/Z_{n+1}$. In this way we reduce to proving the result for F_n in case $Y_{n+1} = Y$, i.e., we may assume that $\mathcal{F}$ is flat in dimensions $\geq n+1$ over Y.

Fix n and assume $\mathcal{F}$ is flat in dimensions $\geq n+1$ over the affine scheme Y. To finish the proof we have to show that F_n is representable by a monomorphism $Z_n \to S$ of finite presentation. Since the question is local in the étale topology on Y it suffices to show that for every $y \in Y$ there exists an étale neighbourhood $(Y', y') \to (Y, y)$ such that the result holds after base change to Y'. Thus by Lemma 05HU we may assume there exist étale morphisms $h_j : W_j \to X$, $j = 1, \ldots, m$ such that for each j there exists a complete dévissage of $\mathcal{F}_j/W_j/Y$ over y, where $\mathcal{F}_j$ is the pullback of $\mathcal{F}$ to W_j and such that $|X_y| \subset \bigcup h_j(W_j)$. Since h_j is étale, by Lemma 0CWF

the sheaves $\mathcal{F}_j$ are still flat over in dimensions $\geq n+1$ over Y. Set $W = \bigcup h_j(W_j)$, which is a quasi-compact open of X. As $\mathcal{F}$ is pure along X_y we see that

$$E = \{t \in |Y| : \mathrm{Ass}_{X_t}(\mathcal{F}_t) \subset W\}.$$

contains all generalizations of y. By Divisors on Spaces, Lemma 0CVV E is a constructible subset of Y. We have seen that $\mathrm{Spec}(\mathcal{O}_{Y,y}) \subset E$. By Morphisms, Lemma 05LW we see that E contains an open neighbourhood of y. Hence after shrinking Y we may assume that $E = Y$. It follows from Lemma 05UD that it suffices to prove the lemma for the functor F_n associated to $X = \coprod W_j$ and $\mathcal{F} = \coprod \mathcal{F}_j$. If $F_{j,n}$ denotes the functor for $W_j \to Y$ and the sheaf $\mathcal{F}_j$ we see that $F_n = \prod F_{j,n}$. Hence it suffices to prove each $F_{j,n}$ is representable by some monomorphism $Z_{j,n} \to Y$ of finite presentation, since then

$$Z_n = Z_{1,n} \times_Y \ldots \times_Y Z_{m,n}$$

Thus we have reduced the theorem to the special case handled in More on Flatness, Lemma 05UF. $\square$